Lecture 6: Forecast Communication, Monitoring, and Governance

ECON 417 Business Forecasting

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How Lecture 6 Fits the Course

- Lectures 1–5 taught us how to build, validate, stabilize, and quantify forecasts.
- Lecture 6 asks: how do we manage forecasts after they leave the notebook?
- This is the bridge from classroom forecasting to organizational forecasting practice.

Key message: A model is not complete until someone knows how to monitor, update, and act on it.

Agenda (Today)

1. Communication
2. Monitoring
3. Drift and Governance
4. From Statistical Error to Business Loss
5. Conceptual Introduction to Deep Learning
6. Governance Checklist

Forecast Communication Is Part of the Forecast

Definition: Forecast communication

Forecast communication means translating model output into language, visuals, and action rules that a decision-maker can use without understanding the model internals.

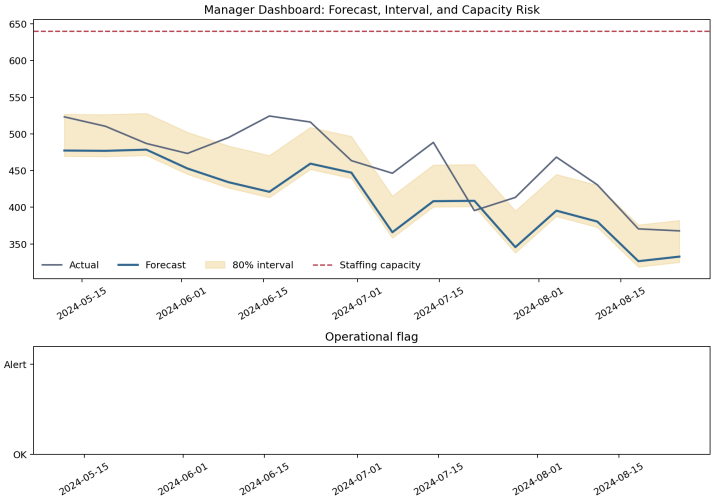
- Decision-makers need three things: the expected path, the uncertainty range, and the action trigger.
- A technically correct model can still fail if the communication is unusable.

Key message: The forecast is not the final product. The decision is.

Notebook Example: COVID Crash Manager Dashboard (Example 6.1)

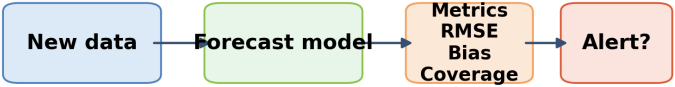
- Dashboard covers January–May 2020, the COVID crash window.
- Shows champion model daily SP500 forecasts, realized returns, 80% intervals, and risk flags for days where the lower bound breaches the drawdown threshold.
- Designed so a non-technical risk manager can immediately identify which days required action.

Notebook Example 6.1: Manager Dashboard (Jan–May 2020)



Monitoring Deployed Forecasts

Monitoring asks whether the deployed model is still behaving well



If metrics cross a threshold, investigate drift, retrain, recalibrate, or escalate.

concept adapted from: IBM model drift

Which Metrics Should We Track?

Definition: Monitoring

Monitoring means checking whether forecast accuracy, bias, and uncertainty calibration remain acceptable after the model is deployed in production.

- **Rolling RMSE**: are forecast errors growing?
- **Rolling bias**: are forecasts systematically too high or too low?
- **Interval coverage**: are uncertainty bands still honest?

Practical rule

Define alert thresholds before failure happens, not after.

Notebook Example: Rolling Monitoring Metrics (Example 6.2)

- 60-day rolling RMSE, rolling bias, and rolling 80% coverage for the champion model deployed from January 2020.
- Monitoring charts show a sharp deterioration during the COVID crash: rising RMSE, persistent negative bias, and collapsing coverage.
- The failure was visible in the metrics within weeks of the crash onset—before any scheduled review date.

Notebook Example 6.2: Rolling Monitoring Metrics

What Is Drift?

Definition: Data drift

Data drift means the distribution of input features shifts away from what the model was trained on, so the model operates in unfamiliar territory.

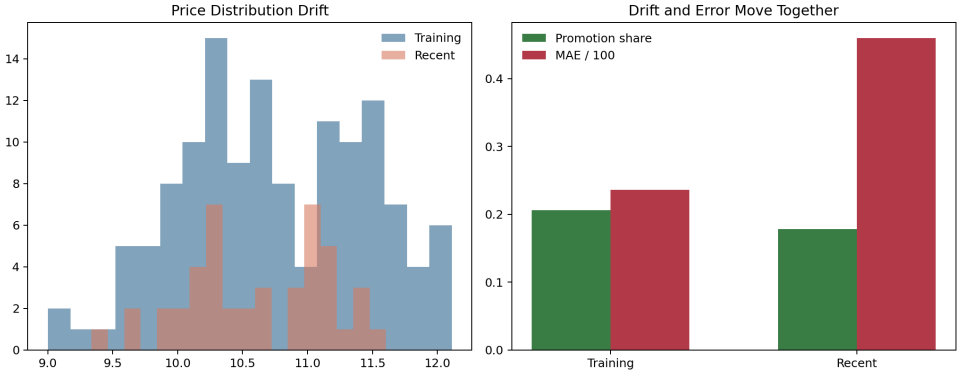
Definition: Concept drift

Concept drift means the relationship between inputs and the target changes, so even correctly observed inputs lead to wrong predictions.

- **Example (data drift):** VIX spikes into a range never seen during training.
- **Example (concept drift):** VIX's predictive relationship to SP500 returns breaks down during a market regime shift.

Key message: If the environment changes, yesterday's model can become today's liability.

Drift Detection in Practice



monitoring concept supported by: GitHub / DriftGuard

Notebook Example: COVID as Governance Stress Test (Example 6.3)

- Four panels: VIX distributions (training vs. COVID vs. post-COVID), VIX time series with regime shading, rolling VIX coefficient (concept drift indicator), monitoring RMSE exceeding alert threshold.
- Both data drift (VIX far above training range) and concept drift (changed sensitivity) were detectable signals before the worst model failures.

Key message: Define drift triggers before the crisis, not after.

Champion vs Challenger

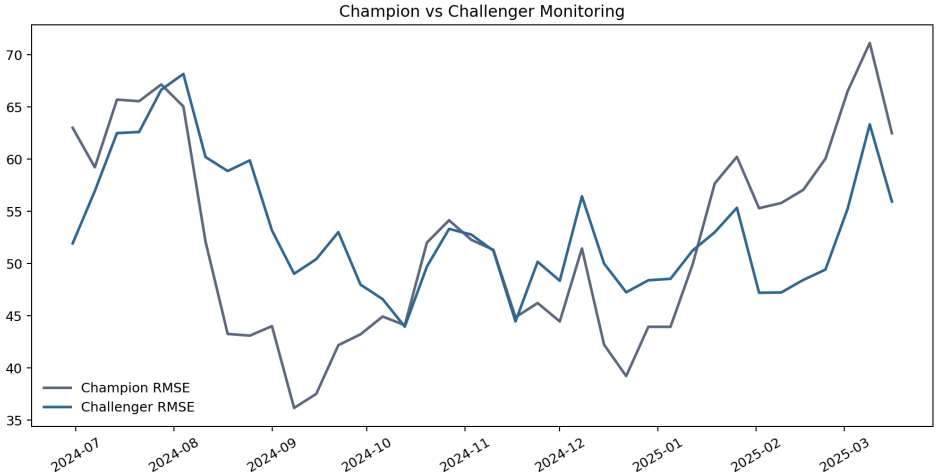
Definition: Champion-challenger framework

The champion is the model currently in production; the challenger is a competing model tracked in parallel to gather evidence before any replacement decision.

- Replacement should depend on sustained evidence across many evaluation windows.
- Changing models too often creates operational noise.
- Refusing to compare alternatives can leave the firm with a deteriorating process.

Key message: Governance means making model changes deliberately, not impulsively.

Notebook Example: Ridge Champion vs. RF Challenger (Example 6.4)

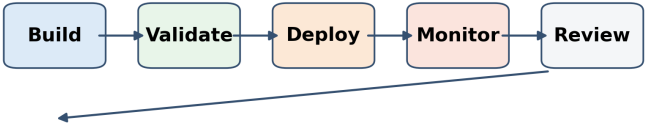


Notebook Example: Champion vs. Challenger Details

- Champion: Ridge regression trained on 2015–2019.
- Challenger: Random Forest, same training window.
- 60-day rolling evaluation in the 2020–2024 deployment period.
- RF challenger wins a majority of rolling windows—especially during COVID—providing statistical grounds for a model transition conversation.

Governance Lifecycle

Governance makes forecasting reproducible, auditable, and accountable



Documentation + ownership + thresholds + override logs

concept adapted from: IBM model drift monitoring

Human Overrides and Accountability

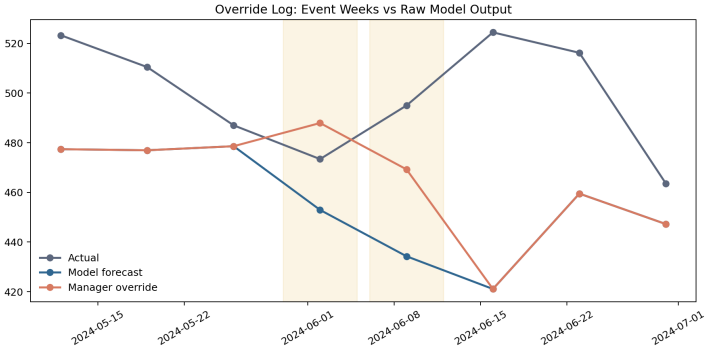
Definition: Forecast override

A forecast override is a human adjustment made after seeing the model output, usually because of new business information not captured in the historical data.

- Overrides are allowed when new information genuinely matters.
- But overrides must be logged, reasoned, and reviewed later.
- This protects against hindsight bias and keeps decision responsibility visible.

Key message: A good governance system does not eliminate judgment—it records judgment.

Notebook Example: COVID Override Audit Trail (Example 6.5)



- Five simulated interventions during the COVID monitoring period.
- Each override tagged with forecast, override value, realized outcome, and correct/incorrect label.

Business Loss Function

Definition: Business loss function

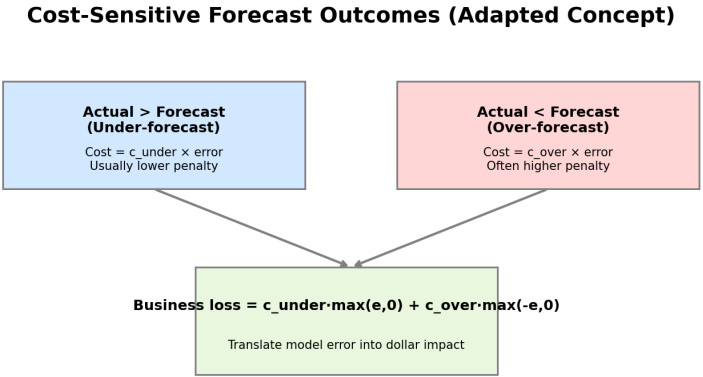
A business loss function assigns a dollar cost or operational penalty to each unit of forecast error, making the link between statistical accuracy and organizational impact explicit.

$$\text{Business Loss}_t = c_{\text{under}} \cdot \max(e_t, 0) + c_{\text{over}} \cdot \max(-e_t, 0)$$

$$\overline{\text{Loss}} = \frac{1}{T} \sum_{t=1}^{T} \text{Business Loss}_t$$

- RMSE measures error in return units; business loss translates error into dollars.
- A 5 bp forecast miss at maximum leverage can breach risk limits and trigger forced liquidation.
- Governance conversations should reference dollar loss, not just RMSE.

Business Loss: Concept

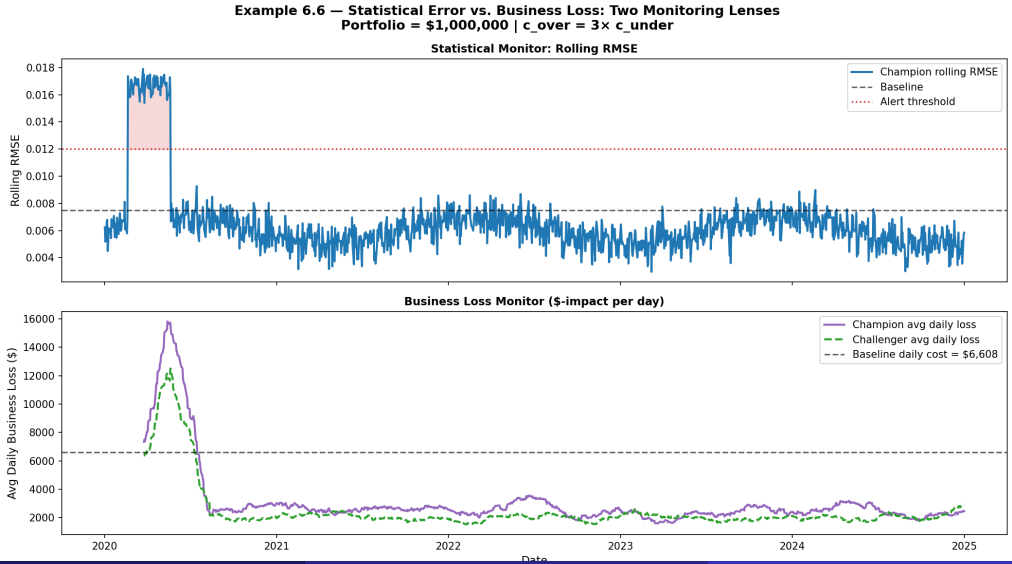


concept adapted from: Towards Data Science – cost-sensitive prediction

Notebook Example: $1M Portfolio Business Loss (Example 6.6)

- Asymmetric loss ($c_{\text{over}} = 3$, $c_{\text{under}} = 1$) applied to a $1,000,000 simulated equity portfolio.
- Each day's forecast error is translated into a dollar cost for both Ridge champion and RF challenger.
- 60-day rolling average cost shows the RF challenger incurs meaningfully lower business loss during the COVID crash window.
- This dollars-and-cents argument complements the RMSE comparison for the model transition decision.

Notebook Example 6.6: Rolling Business Loss – Champion vs. Challenger



LSTM and Deep Learning for Forecasting

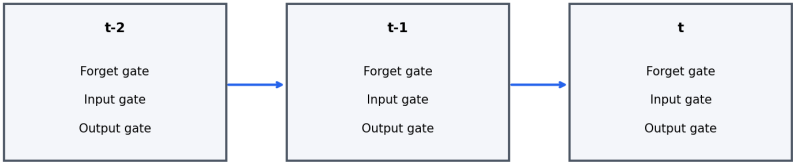
Definition: LSTM (Long Short-Term Memory)

An LSTM is a type of recurrent neural network designed to learn patterns from long sequences, capturing dependencies across many time steps without requiring the analyst to manually specify AR orders or interaction terms.

- LSTMs maintain a cell state that can carry information across hundreds of time steps.
- Three gates (forget, input, output) control what information is retained or discarded.
- Appealing for long, complex time series where the relevant memory depth is unclear.

LSTM Architecture: Concept

LSTM Cell Logic (Adapted Concept)



Cell state carries long memory across time

Gates decide what to forget, update, and reveal

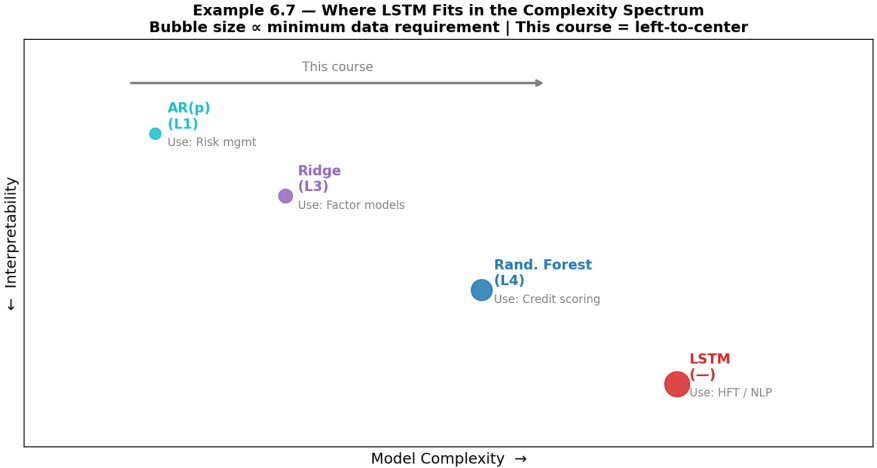
concept adapted from: IBM – What is LSTM?

How LSTM Differs from Methods in This Course

Feature	L1–L6 methods	LSTM
Interpretability	High (coefficients, rules)	Low (millions of weights)
Data requirement	Works on small samples	Needs large datasets
Validation discipline	Train/val/test split	Identical—same rules apply
Forecast intervals	Residual quantiles	Requires additional techniques
Governance / auditability	Straightforward	More complex

Key message: The validation discipline never changes. LSTM does not exempt you from train/val/test.

Notebook Example: Complexity vs. Interpretability Positioning (Example 6.7)



Notebook Example 6.7: Positioning Takeaways

- OLS, Ridge, RF, and LSTM positioned by data requirement, interpretability, and governance cost.
- For daily-return forecasting on modest datasets, simpler models typically perform comparably.

When to Consider Deep Learning

1. Start with the simplest model that beats baselines out of sample.
2. Add complexity only when validation results clearly justify it.
3. Use LSTM only when the dataset is large, simpler models leave substantial unexplained structure, and the deployment context tolerates lower interpretability.

Regulatory reality

In regulated settings (banking, insurance, public agencies), model explainability is often required by policy. A model that cannot be explained to a non-technical committee is difficult to deploy regardless of its predictive accuracy.

Five Questions Every Forecaster Must Answer

Before declaring a model ready for production:

1. **What data were used?** Source, time range, preprocessing, known quality issues.
2. **What validation rule was followed?** Chronological split; train/val/test proportions; no look-ahead.
3. **What uncertainty measure was reported?** Interval width, coverage rate, residual distribution check.
4. **What monitoring threshold was chosen?** The rolling RMSE or bias level that triggers a review.
5. **What would trigger re-estimation?** A specific drift signal, coverage drop, regime event, or calendar date.

Key message: A model is not complete until its maintenance plan is also clear.

Notebook Example: Governance Checklist Printout (Example 6.8)

- Governance checklist evaluated for the champion model: trained 2015–2019, deployed Jan 2020.
- Reveals specific weaknesses: no pre-specified drift trigger for a VIX regime shift, and no coverage monitoring threshold.
- Gives students a concrete template applicable to any forecasting project.

Takeaways

1. Good forecasting includes communication, not just modeling.
2. Deployed models need continuous monitoring for error, bias, and coverage.
3. Data drift and concept drift require predefined triggers and escalation paths.
4. Business loss in dollar terms is more persuasive to decision-makers than RMSE.
5. Champion-challenger evaluation makes model transitions deliberate and evidence-based.
6. LSTM and deep learning require the same validation discipline—they are not exempt.
7. Governance makes forecasting reproducible, auditable, and safe for real decisions.

Key message: A forecast becomes professional when it can be explained, monitored, and maintained.